MIKHAIL CHULOSHNIKOV

Senior Backend Engineer | Principal Architect | Team Lead | CTO | High-Load & Systems Researcher

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TECHNICAL STACK

Languages

PHP, Go, Python, TypeScript

Common

Symfony, Doctrine, Vue, Angular, Capacitor, MySQL, PostgreSQL, Redis, Memcached, RabbitMQ, Kafka, ClickHouse, Cassandra, Ansible, Docker, Git, CI/CD, Nginx, Caddy, Raw sockets, Github Actions, Debian, Elasticsearch, Grafana, Loki, Alloy, Sharding, Load Balancing, SOA, Exploit, RCE, Buffer Overflow, OllyDbg, GDB, Deobfuscation, Strace

EXPERTISE

- Core Banking & Payments
- High-Load
- Reverse Engineering
- Vulnerability Research
- Team Leadership

CERTIFICATIONS

Zend PHP Certified Engineer

EDUCATION

Synergy University
Fullstack Dev • 2026

Modern Technical Inst.
Service • 2015

Moscow College of Math
Programming & AS • 2008

17+
YEARS OF EXPERIENCE

I bridge the gap between low-level system engineering and scalable web architecture. Leveraging 17+ years of experience, I build mission-critical backend ecosystems where performance, security, and stability are solved at the foundation.

SELECTED IMPACT CASES

Silverbird (Theropod LTD)

Oct 2022 — March 2024

Project Lead / Senior Developer

- **Core Banking Development:** Engineering of core banking engines and all related financial infrastructure.
- **Multi-PSP Integration:** Design and development of systems for seamless multi-PSP (Payment Service Provider) integration.
- **Payment Processing Core:** Architecture and development of the core engine for Inbound/Outbound payments, FX transactions, and intelligent payment routing.
- **Leadership & Stakeholding:** Acting as Team Lead and Stakeholder for the core banking and payment modules.
- **External Ecosystems:** Integration of external PSPs and development of specialized Beneficiary Modules.
- **Security Auditing:** Conducting comprehensive security audits for financial components.
- **Banking Domain Expertise:** Full-cycle implementation of payments, FX, and transactions. Deep integration with payment schemes: SWIFT, SEPA, Faster Payments, T2, and platforms like TUUM / Integrated Finance.

FINTECH INFRASTRUCTURE | PAYMENTS PROCESSING | SECURITY AUDIT

TSUM (Mercury Group)

April 2021 — April 2022

Team Lead

- **Cross-functional Leadership:** Managed a complete product cell including Developers, QA, Product Managers, and Business Analysts.
- **Massive Scale:** Managed High-Load systems handling nearly 1 Billion data records.
- **Service Ownership:** Full accountability for 6+ mission-critical services within the e-commerce ecosystem.
- **Marketing-Tech Strategy:** Spearheaded the Marketing product-engineering bridge, acting as a primary Stakeholder.
- **Team Transformation:** Completely rebuilt and upskilled the team to align with modern product requirements and tech standards.
- **Innovation Driver:** Initiated and oversaw the development of new services using cutting-edge technologies to solve specific business needs.
- **Process Excellence:** Built internal development workflows from scratch and streamlined interaction protocols between business units and external teams.

HIGH-LOAD (1B+ RECORDS) | MANAGEMENT | STRATEGIC PLANNING

MIF

Sept 2008 — Aug 2009

Security & Low-Level Systems Researcher

- **Full-Chain Vulnerability Analysis:** Conducted end-to-end security research on major browser engines and Adobe Reader. Developed functional RCE (Remote Code Execution) exploits (Only Proof-of-Concept) to demonstrate critical memory corruption flaws using WinDbg, x86 Assembly, and advanced heap-spraying techniques.
- **High-Concurrency C&C Architecture:** Engineered ultra-lightweight remote management agents (4KB binaries) and command-and-control (C2) infrastructures using C (WinAPI) and Lua. Focus on extreme binary optimization and stealthy network communication.
- **Polymorphic Code Protection:** Developed a sophisticated machine-based obfuscator. It featured a custom code translator (Python/JS to pseudocode) designed to bypass heuristic analysis and static signature detection engines.
- **Algorithmic Network Topology:** Designed and implemented automated systems for large-scale content generation and inter-linked network structures (Ring & Pyramid topologies). Optimized link-graph algorithms to maximize search engine indexing efficiency.
- **Infrastructure Monetization:** Developed specialized platforms for client hardware resource acquisition and satellite site generation systems.
- **Systems Mindset:** This era defined my "low-level first" approach. Understanding memory corruption and CPU-level execution is what allows me today to build high-load systems that are inherently secure.

REVERSE ENGINEERING | X86 ASSEMBLER | MEMORY CORRUPTION

CURRENT VENTURE

Alunea

2025 — Present

Founder & Lead Architect • alunea.io

An AI-driven language learning ecosystem that bridges passive content consumption with active speaking practice using **Voice-Interactive Tutors** and real-time subtitle processing.

- **AI Tutor "Violet":** Interactive voice calls for active recall using context-aware vocabulary.
- **Smart Subtitles:** Contextual dual subtitles for Netflix & YouTube with on-the-fly generation.
- **Unified Sync:** Real-time word-sync across Browser Extensions, iOS/Android apps, and Backend.

LLM INTEGRATION | VOICE AI | ADAPTIVE LEARNING | CROSS-PLATFORM SYNC

PROFESSIONAL CHRONOLOGY

JULY 2024 — OCT 2025

Aventus IT | Team Lead

Leading/mentoring 7 backend devs. Streamlining internal workflows and cross-departmental operations. Designing robust architectures for high-performance services. CI/CD optimization. Direct contribution to flagship microservices.

OCT 2022 — MARCH 2024

Silverbird (Theropod LTD) | Project Lead / Senior Dev

Engineering of core banking engines, multi-PSP integration, and FX payment processing. Leadership for payment modules. Integration with SWIFT, SEPA, Faster Payments, T2, TUUM. Comprehensive security auditing.

JULY 2022 — SEPT 2022

MTS | Senior Developer

Engineering and implementation of new functional modules within the Bitrix framework. Architectural migration of legacy Bitrix functionality to modern microservices.

APRIL 2021 — APRIL 2022

TSUM (Mercury Group) | Team Lead

Managed product cell (Dev, QA, PM, BA). Ownership of 6+ mission-critical services handling 1B+ records. Spearheaded Marketing-Tech strategy. Rebuilt and upskilled the team.

SEPT 2020 — APRIL 2021

Silverbird | Head of Development

Orchestrated end-to-end creation of banking platform. Assembled engineering team from scratch. Designed and launched ~10 projects (API, microservices, infra). Managed technical governance and global architecture.

SEPT 2020 — APRIL 2021

Gosuslugi | Tech Analyst / Senior Dev

Development of core feedback subsystem. Profiling and optimization for extreme high-load. Complex integration with ESIA (National SSO) and ROI analytical systems. Data normalization.

FEB 2018 — JULY 2020

TSUM (Mercury Group) | Senior Developer

Engineering of core internal ecosystem. Debugging complex data synchronization. Implementation of high-performance search and MarTech services (feed generators, QR engines).

APRIL 2016 — NOV 2017

JivoChat (JivoSite) | Team Lead

Managed team of 6. Implemented Scrum, boosting productivity from 70% to 120%. Automated systems for transparency. Owned full release management cycle.

JAN 2016 — MARCH 2016

VK | Senior Developer

High-load specialized projects (NDA).

AUG 2014 — DEC 2015

Activizm.ru | Senior Developer

AdTech integration (Google AdWords/Yandex.Direct API), large-scale data exports to Merchant Centers, and system profiling.

NOV 2013 — JULY 2014

Mamba | Senior Developer

Developed notification engine for tens of millions of users. Integration with custom C-based daemons. Refactoring of mail gateway protocols. Database optimization (SQL, RRDtool).

MAY 2012 — AUG 2013

Enter.ru | Senior Developer

Fintech integration for online credit purchases, 1C integration, Symfony migration.

AUG 2009 — APR 2012

Wikimart | Senior Developer

Billing system from scratch, high-throughput ETL for massive product feeds.

SEPT 2008 — AUG 2009

MIF | Systems Researcher

Vulnerability research (RCE PoCs), WinDbg, x86 Assembly. Engineering 4KB C/WinAPI remote agents. Polymorphic code obfuscation. Algorithmic network topology optimization.

PUBLICATIONS

SECURITY

Buffer Overflow for Beginners

An in-depth look at memory corruption and stack smashing.

NETWORKS

The Fundamental Vulnerability: ARP Spoofing

Analyzing low-level network protocol flaws and mitm attacks.

SYSTEMS

The Power of UDF Functions in MySQL: Data Encryption Using C++ and OpenSSL

Implementing custom UDF functions for high-performance crypto.

STORAGE

File-based hash map — using as persistent data storage

Designing a high-performance hash map for disk-based data.

CODE

C / C++

- C/C++/Golang: Algorithms
- C++: MySQL OpenSSL Plugin
- C++: ARP Spoofer
- C++: Shellcode Generator
- C++: YansDB (NoSQL Storage)
- C++: TCP/IP Broadcast Server
- C++: Lightness HID Tool

Python, Go, Lua

- Go: LSM-tree based storage
- Go: Concurrent Crawler
- Python: Helicopter Game
- Python: L2 Lottery Bot
- Python: Web Cryptor
- Python: Lightness Monitor
- Lua: Simple Installer
- Godot: Fox Wild Hunt

PHP, Java, TypeScript

- PHP: File Hash Map
- PHP: Worker Pool (System V)
- Java: Email Alias - Spring Backend
- TypeScript: Email Alias - Angular front
- Python: Email Alias - Ansible CI/CD
- PHP: Various PHP Snippets

BROAD STACK

Note: *While not all of these are daily drivers, I possess the practical experience to implement, debug, or lead projects using this broader ecosystem.*

LANGUAGES & CORE

PHP / Go / Java

Symfony • Laravel • Yii • Zend • Custom Extensions (C) • Doctrine • Libs • High-perf Microservices • Concurrency • Spring Security/Boot/Social • Hibernate • Apache JBI/ESB/JMS/Service Mix/ActiveMQ/Camel

Python / TypeScript / Lua / Gdscript (Godot)

Django • SQLAlchemy • Vue • Vuetify •

SYSTEMS & HIGH-LOAD

Databases

PostgreSQL • MySQL • MS SQL • Redis • Memcached • ClickHouse • Cassandra • Interbase • Firebase • Kyoto • Hash • Cardinality • Constraints • Replication • Sharding • Partitioning • Performance Tuning • Profiling • Locks • Indexes • T-SQL-92

CI/CD

Docker • Ansible • Github Actions

SECURITY & LOW-LEVEL

Sec Research & Audits

0-day Exploit Dev • RCE • XSS • Buffer Overflow • Cryptors • Shellcode

Reverse Engineering

OllyDbg • WinDbg • GDB • Binary Patching • Deobfuscation • Import Table Recovery

Deep Debugging

Strace Expert • Kernel Module Patching

Angular • Electron/Ionic/Capacitor • Webpack • Lua Stdl/COM/fs/socket/C API

C / C++ / ASM / TCL / Delphi / VBA / Bash

Boost • WinAPI/POSIX • Kernel • Stdl • Raw sockets • COM • Stack • Heap • Registers

Infrastructure

Debian • Nginx • Caddy • Apache • Tomcat • Jetty • Elasticsearch • Sphinx • Solr • RabbitMQ • Kafka • Grafana • Loki • Alloy • Git • Svn • Mercurial • Jira • Confluence • Trac • Redmine

Architecture

Sharding • Load Balancing • Zero-Downtime Altering • MSOA • Distributed Systems • Raw Sockets • Packet Inspection (TCP/UDP)

THE BACKSTORY: FROM DELPHI TO ARCHITECT

GENESIS & EARLY EXPLOITS

Started at 14 with a Delphi book and no internet. By college, I was hacking WinServer networks and maintaining **WoW servers (MaNGOS)** on C++, learning the grit of binary patching. In the **Lineage 2** era, I built a custom Socks4/5 proxy layer for on-the-fly packet modification — pure raw socket magic invisible to anti-cheats.

THE "GREY" ERA & OPTIMIZATION

Building botnets taught me extreme optimization: I integrated a **Lua interpreter into a C-core**, feeding it raw bytecode to achieve a **4KB binary**. Developed custom cryptors using Python-to-Pseudocode translators and bypassed top-tier AVs by shifting from standard WinAPI to CMD/FTP shellcode injection.

ARCHITECTURAL PHILOSOPHY & "RELIGION"

- **Infra:** Debian only. Use **UDP** instead of TCP for real-time performance.
- **DB:** Foreign Keys are for billing only. Stop the subquery madness and DB-trash-can approach.
- **Arch:** Thin clients (PHP) → High-speed Microservices (Go/Java). Decentralized P2P trust rings.
- **Logic:** Debugged Pinba stack corruption for 24h. I don't trust "magic" monitoring tools anymore.

"I love computers, I understand them, and they understand me."